

models with additional knowledge data. These experiments aim to validate the causal relationship between improved parameter specialization and enhanced model performance on knowledge tasks.

5.1 Finetuning Setup

We randomly selected 10 high-frequency concepts from the SpecWiki benchmark. For each concept, we gathered relevant textual material from the top 10 most popular Google search results, including the corresponding Wikipedia article, and compiled this into an additional finetuning training dataset.

Next, we will validate whether the improvement in Parameter Specialization and the enhanced efficiency in the model’s use of knowledge truly exhibit a causal relationship through four distinct finetuning experiments. The experimental setups are detailed below:

FT-FV(Full Vectors) Perform full finetuning (FT) on all parameter vectors of the MLPs across all layers in the model, while keeping the other parameters frozen.

FT-PV(Partial Vectors) Perform partial finetuning on a subset of the parameter vectors in the MLPs while keeping the other parameters frozen. Specifically, for each model, we apply finetuning (FT) to the top $\frac{k}{8}$ most highly activated parameter vectors**. For the selection of k , please refer to the description in §4.1.

FT-CV(Complementary Vectors) Perform finetuning only on the complementary set of parameter vectors, excluding the target vectors.

FT-RV(Random Vectors) Perform finetuning on a subset of parameter vectors randomly selected from the MLP, ensuring the same quantity as in the FT-PV setting.

5.2 Finetuning Results

In addition to evaluating the model’s performance on SpecWiki’s Multi-choice Question and Open-ended Generation tests, as well as the Parameter Specialization scores, we also report two other metrics, Semantic Entropy (Kuhn et al., 2022) and Local Intrinsic Dimension (LID) (Yin et al., 2024), for measuring the extent of hallucination in the model’s output. These metrics help evaluate whether training strategies that enhance Parameter Specialization—by aligning better with the model’s knowledge retrieval mechanisms through a data-encoded strategy—can effectively reduce the unintended side effect of hallucination. For a detailed introduction to these two hallucination measurements, please refer to §B.3.

The final results are presented in Table 2. From the results, we can see that the FT-PV method, which finetunes only a small subset of the highly activated knowledge parameters, not only further enhances the model’s Parameter Specialization compared to the three other finetuning setups, but also greatly improves the model’s utilization of specific knowledge. As a result, it achieves the best performance on the benchmark Multi-Choice questions and Open-ended generation tasks. Additionally, by reducing the influence of irrelevant information in the model’s key parameter vectors, FT-PV helps to significantly reduce the level of hallucination in the generated text.

Although FT-FV and FT-CV does improve the model’s performance on both the Multi-Choice questions and Open-ended generation tasks to some extent, compared to FT-PV, it does not lead to a better increase in Parameter Specialization. Additionally, the degree of hallucination in the generated text is not effectively reduced. FT-RV, serving as a counterpart to FT-PV, demonstrates that fine-tuning the same number of arbitrary value vectors in the model’s MLP can not result in a desirable knowledge enhancement.

6 Conclusion

This study reveals that enhanced parameter specialization—where related knowledge is encoded in focused parameter vectors—correlates with superior performance in large language models. Analyzing 20 open-source models, we observed stronger models increasingly consolidate similar knowledge into fewer parameters, while weaker models distribute it diffusely. Controlled experiments confirmed

**We experimented with $\frac{k}{2}$, $\frac{k}{4}$, $\frac{k}{8}$, and $\frac{k}{16}$, and found that finetuning only $\frac{k}{8}$ of the parameter vectors was sufficient to achieve excellent performance.

that optimizing this specialization improves task performance and reduces hallucination. These findings highlight the importance of aligning knowledge storage with models’ retrieval mechanisms for efficiency and accuracy. Future work should explore dynamic knowledge updates and scalability, advancing both interpretability and performance in LLM design.

7 Limitations and Future Work

In our work, we have only examined and validated knowledge parameter specialization within the MLP, and this was done by treating vectors in the MLP as units of analysis. However, at least this remains one of the knowledge storage methods that has been extensively validated so far (Geva et al., 2021a; Meng et al., 2022; Geva et al., 2023). In fact, knowledge may also reside within the attention module of transformer models (Geva et al., 2023).

Additionally, due to GPU limitations, all the models we tested are smaller than or equal to 14B parameters, so we were unable to validate our conclusions on larger models, such as those with 35B parameters or more.

In future work, we will progressively narrow the focus of our research to individual neurons in language models, aiming to measure and validate more precise Parameter Specialization and Parameter Superposition. In addition, we will extend this concept to the study of other related model architectures, including Mixture of Experts (Fedus et al., 2022), which similarly enhances model performance by specializing expert parameters, as well as Sparse Auto-Encoders (Huben et al., 2024), which help clarify the model’s representations by leveraging a larger parameter space and mitigating the superposition of these features.

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